

HW04 — Automation Testing — Main Report

Student: Nguyễn Thành Dâng (23127334) **Repository:** <https://github.com/ThanhDang-Vn/software-testing> **Demo video (Agent Skill + end-to-end flow):** https://youtu.be/e-_aoQkVflk **System under test:** EShop (React web frontend + Node/Express + SQLite backend) **Features under test:** FR-02 Login & Account Lockout (Pool A), FR-11 Order History (Pool B), FR-14 Category Management CRUD (Pool C)

1. Overview

This report documents an AI-assisted, human-reviewed automation suite for three EShop features, one from each required pool. The suite is data-driven, runs on three real browser engines, and is executed end to end to produce multi-browser HTML reports. Where a failing assertion exposed a genuine product defect, the defect is reported here and published as a GitHub Issue with evidence.

The work follows an *AI-first, human-verified* model: an AI tool proposed the initial scripts and data; the student then reviewed each proposal against the requirement, the SUT source, and the execution evidence, correcting or rejecting anything that was fragile, incomplete, or wrong. Every AI interaction and the corrections applied to it are recorded in `documents/ai-audit-report.md`.

2. Tools and environment

Area	Choice
Language	TypeScript (ESM)
Test runner	@playwright/test 1.55
Runtime	Node.js 20
Browser engines	Chromium, Firefox, WebKit (three Playwright projects)
Reporter	Playwright HTML reporter (+ JSON for the run manifest)
Data format	External JSON case files; credentials injected via environment variables
AI tool	Declared in the AI Audit Report

Every HTML report title embeds Run by: 23127334 together with an ISO 8601 timestamp so the run is attributable, as required by the anti-cheat constraints.

3. Test design and data-driven approach

The suite contains **44 logical cases** across the three features:

Feature	Cases	Composition
FR-02 Login & Lockout	15	positive login, negative auth, field validation, UI contract, lockout state/boundary/edge
FR-11 Order History	13	7 UI presentation + 6 backend API (authorization / cross-user ownership)
FR-14 Category CRUD	16	6 UI + 10 backend API (create / view / delete, validation, role authorization)

Data-driven design. Case metadata (IDs, categories, expected outcomes) lives in external JSON under `automation/test-data/`, decoupled from the spec logic. A typed loader validates the data on read: it rejects missing or duplicate IDs, unknown action types, invalid schemas, and any raw credential embedded in the data file. Credentials are supplied only through environment variables, and synthetic users/categories are generated per run with a unique run ID so parallel or repeated executions never collide.

Isolation and cleanup. FR-14 UI mutations remove the records they create in a `finally` block. FR-11 UI-presentation cases use deterministic mocked fixtures so rendering assertions are stable, while the authorization and ownership cases exercise the real backend.

Assertion patterns. The suite uses at least three distinct patterns:

1. Web-first UI assertions — URL, visibility, text, attribute, CSS class, and element count.
2. Primitive equality on HTTP status codes for the API cases.
3. Structural assertions on arrays/objects, including asymmetric containment on response shapes.

Selector and wait strategy. Selectors prefer accessible roles, placeholders, stable visible text and exact table cells. There is no XPath, positional `nth`, `waitForTimeout`, raw click-via-evaluate, or arbitrary UI delay. Waits are web-first (auto-retrying) except for the single FR-02 lock-expiry case, where the wait is an explicit business window (the account-lock duration), not a UI sync hack. Details in `documents/selector-wait-review.md`.

4. Execution results

Each feature ran on all three engines — **9 feature/browser runs, 132 test-browser executions**.

Feature	Chromium	Firefox	WebKit
FR-02	9 pass / 6 fail	9 / 6	9 / 6
FR-11	12 / 1	12 / 1	12 / 1
FR-14	10 / 6	9 / 7	10 / 6

Totals: 132 executions — **92 passed, 40 failed, 0 skipped.**

The 40 failures are not test errors: they are **repeatable assertion failures that map to confirmed SUT defects** (Section 5), each corroborated by the backend source code and by report/trace artifacts. The single Firefox-only extra failure (FR14-TC-008) was a browser-context teardown event; the case then passed 3/3 in targeted repetition and is classified **FLAKY**, not a product defect.

Evidence: reports/run-manifest.json, the nine reports under reports/html/, and documents/execution-and-stability-review.md.

5. Confirmed defects

Eight distinct defect groups were confirmed and published as GitHub Issues **#33–#40** (labels bug, hw4), each with a screenshot and, for API defects, a reproducible REST request.

ID	Feature / cases	Expected	Actual	Severity
BUG-01	FR-02 TC-009/010 — login input types	email / password input types	both text (password not masked)	P1
BUG-02	FR-02 TC-012/013 — lock arithmetic	+1 per failure; lock on 3rd	+2 per failure; premature lock	P0
BUG-03	FR-02 TC-014/015 — lock duration & feedback	specific message; 30 s window	generic message; 180 s window	P0
BUG-04	FR-11 TC-012 — shipping order	no cancel while shipping	Cancel action shown and accepted	P0
BUG-05	FR-14 TC-005/006/007 — name validation	HTTP 400 on invalid name	HTTP 200, category created	P1
BUG-06	FR-14 TC-009 — delete unknown	HTTP 404	HTTP 200, "deleted"	P1
BUG-07	FR-14 TC-011 — required contract	native/visible required field	neither present	P1
BUG-08	FR-14 TC-016 — authorization	HTTP 403 for customer token	HTTP 200, category created	P0

The most serious findings are security/integrity defects: a broken account-lock rule (BUG-02/03), an order-integrity gap allowing cancellation of a shipping order (BUG-04), and a missing authorization check that lets a customer token create categories (BUG-08). Full reproduction steps and screenshots are in documents/bug-report.md; API cases are reproducible via automation/rest/fr14-bug-evidence.rest.

6. Human review and gap analysis

The AI-generated origin scripts were functional but had recurring weaknesses that human review corrected before the results were trusted:

- **FR-02** — origin used positional inputs, a CSS error-class locator, inline credential assumptions, and a fixed sleep. Review moved data to JSON/environment profiles, added field validation, replaced fragile locators with accessible ones, and made the lock window explicit.
- **FR-11** — origin used row indexes and wrongly *accepted* cancellation during shipping. Review fixed the oracle (cancellation must be blocked), replaced indexes with exact order IDs, added cross-user ownership checks, validated fixtures, and awaited route fulfillment.
- **FR-14** — origin treated a confirmation dialog as a hard requirement and lacked successful create/view/delete and customer-role coverage. Review reclassified confirmation as exploratory, added the missing coverage plus unique-data cleanup, and removed arbitrary waits and click-via-evaluate. An initial role locator that mishandled corrupted UI text caused five timeouts and was corrected to a role + stable-prefix locator after trace review.

Why the AI missed these: the model optimized for a script that runs, not for oracle correctness or requirement fidelity — so it accepted the SUT's behavior as the expected result (e.g. shipping cancellation), leaned on positional/implementation-coupled locators, and under-covered negative and authorization paths that the prose requirement implied but did not spell out.

Work left to humans (documented, not automated):

- FR-14 *Update* behavior is a requirement ambiguity: the reviewed acceptance criteria list create/view/delete while the title says "CRUD". This is recorded rather than silently assumed.
- Confirmation-dialog usability and broad visual quality are exploratory and not FR-14 pass/fail criteria.

7. Agent Skill

A reusable Agent Skill packages this workflow — data-driven, multi-browser script generation and evidence capture — so it can be re-applied to new features. The skill and its end-to-end usage are demonstrated in the linked demo video.

8. Self-assessment and conclusion

Criterion	Max	Self-assessed
Task 1 — FR-02	25	25
Task 1 — FR-11	25	25
Task 1 — FR-14	25	25
Task 2 — Demo video	15	15

Criterion	Max	Self-assessed
Agent Skill	10	10
Total	100	100

The suite meets the core Task 1 requirements: three pooled features, 44 data-driven cases, three assertion patterns, three-engine execution with attributable HTML reports, a human-reviewed script set, and eight confirmed defects published with evidence. The failing assertions are genuine product findings rather than test defects, which is the intended outcome of the exercise. The self-assessed grade of 100/100 was selected by the student, who remains responsible for the final scripts and for reviewing every AI-proposed change.